

Individual Notebook Workflow Guide

Football Top 5 Leagues Project - Data Cleaning and Data Analysis Practice

Purpose

Creating one individual practice notebook per team member is a good idea. It allows everyone to practice data cleaning and exploratory data analysis independently before the team merges the best decisions into one official project notebook. This guide explains how each member should work individually while keeping the process consistent, comparable, and easy to integrate later.

Team Rule

Each member should work from the same raw dataset, follow the same cleaning checklist, and document every decision in Markdown. The individual notebooks are for practice and exploration. The final project should use one agreed version of the cleaned dataset.

Recommended Individual Notebook Structure

Section	What to include
1. Introduction	Briefly explain the dataset and the assigned research question.
2. Import Libraries	pandas, numpy, matplotlib, seaborn if allowed, sqlite3 if needed.
3. Load Dataset	Read the CSV and confirm the file loads correctly.
4. Initial Exploration	head(), shape, info(), columns, dtypes, describe().
5. Data Quality Checks	Missing values, duplicates, invalid values, categorical values.
6. Cleaning Decisions	Keep/drop columns, fix types, handle missing values, document choices.
7. Feature Engineering	Create match_result, goal_difference, total_goals, total_xg, xg_difference.
8. EDA Practice	Univariate and bivariate analysis related to the assigned question.
9. Insights	Write 3-5 clear observations from the analysis.
10. Final Notes	What worked, what should be included in the final team notebook.

Step-by-Step Individual Workflow

Step 1 - Start with a clean notebook

Create a notebook named with your role or research question, for example: member_1_xg_analysis.ipynb. Add a title, your name, and the research question you are practicing.

Step 2 - Load the dataset

Import the necessary libraries and load the raw CSV. Confirm that the number of rows and columns matches the original dataset.

Step 3 - Explore before cleaning

Run head(), shape, info(), dtypes, nunique(), and describe(include='all'). Do not clean before understanding the data.

Step 4 - Identify relevant columns

Select the columns related to the four research questions: league, season, goals, xG, current points, previous performance, team ratings, and formations.

Step 5 - Check missing values

Use isnull().sum(). Remember that missing values in previous-performance columns may be normal for early-season matches because teams have not played enough matches yet.

Step 6 - Check duplicates

Use duplicated().sum() for full-row duplicates and check if the match ID is unique.

Step 7 - Fix data types

Convert dates to datetime. Keep goals as integers, xG and ratings as floats, and leagues/formations as categorical variables if useful.

Step 8 - Validate values

Check for impossible values such as negative goals, negative xG, invalid ratings, unexpected league names, or invalid formation codes.

Step 9 - Create shared features

Create the same engineered variables in every notebook: `match_result`, `goal_difference`, `total_goals`, `total_xg`, and `xg_difference`.

Step 10 - Practice EDA for your assigned question

Use summary statistics, `groupby`, visualizations, correlations, and comparisons. Focus only on what helps answer your research question.

Step 11 - Write insights in Markdown

After each important chart or table, write 1-2 sentences explaining what you observe. Avoid leaving charts without interpretation.

Step 12 - Prepare for team integration

At the end of the notebook, list which cleaning decisions, features, charts, and insights should be considered for the final team notebook.

Shared Cleaning Checklist

Check	Method	Decision to document
Dataset overview	head(), shape, info()	What does the dataset contain?
Missing values	isnull().sum()	Drop, fill, or keep? Why?
Duplicates	duplicated().sum()	Remove or keep?
ID uniqueness	df['id'].duplicated().sum()	Is one row one match?
Data types	dtypes	Which columns need conversion?
Dates	pd.to_datetime()	Which date column should be used?
Categorical values	unique(), value_counts()	Are league names/formations consistent?
Invalid values	describe()	Any impossible goals, xG, points or ratings?
Relevant columns	Column selection	Which columns are needed for the questions?
Feature engineering	New calculated columns	Which features should be shared by everyone?

Shared Engineered Features

Feature	Formula / Logic	Useful for
match_result	Home Win / Draw / Away Win based on home_score - away_score	All questions
goal_difference	home_score - away_score	Q1, Q2, Q3
total_goals	home_score + away_score	Q1, Q2, Q3
total_xg	home_xg + away_xg	Q1, Q2, Q3
xg_difference	home_xg - away_xg	Q1
points_difference	home_points - away_points	Q3
recent_form_difference	home recent points - away recent points	Q2
rating_difference	home team rating - away team rating	Q4

Individual EDA Focus by Research Question

Member / Question	Focus	Suggested Analysis
Q1 - xG prediction	xG vs goals and results	Correlation, scatterplots, xG difference vs goal difference, league comparison
Q2 - Recent performance	Previous form vs outcomes	Groupby by form level, previous points vs result, previous xG vs result
Q3 - Current points	Current points vs match performance	Points difference vs result, current points vs goals/xG, league comparison
Q4 - Ratings and formations	Team ratings and formations vs performance	Rating difference vs result, formation comparison, rating difference vs result

Final Team Integration Rules

1. Do not merge all individual notebooks automatically. Review them first.
2. Agree on one official cleaning strategy before creating the final cleaned dataset.
3. Keep only the strongest charts and insights for the final project.
4. Use the same column names and engineered features in the final notebook.
5. Every chart included in the final version must directly support one research question.
6. The final notebook should feel like one project, not four separate assignments.

Recommended Final Output from Each Member

At the end of the individual practice stage, each member should share:

- Their notebook.
- 3-5 important cleaning observations.
- 2-4 useful charts.
- 3-5 written insights.
- Any issues or missing data concerns.
- Suggestions for the final cleaned dataset.